

## Experiment 3: Software Design Principles

### Aim

To identify stakeholders and perform requirement elicitation for the LMS Platform using suitable elicitation techniques, prepare user stories, and develop a Requirement Gathering Report.

### Objectives

- Identify key stakeholders of the LMS Platform.
- Apply requirement elicitation techniques.
- Gather functional and non-functional requirements.
- Prepare user stories with acceptance criteria.
- Document the Requirement Gathering Report.

### Introduction

Requirement elicitation is the process of discovering stakeholder needs and expectations. The quality of software depends on how accurately user requirements are captured. Common elicitation techniques include interviews, questionnaires, observation, workshops and user personas.

### Selected Project

**Project Title:** LMS Platform

**Domain:** Education Technology and Web Application Development

### Stakeholder Identification

| Stakeholder                       | Role                                                   | Requirements                                                                       |
|-----------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------|
| Student                           | Uses the platform for learning, tests and live classes | Easy access to free and paid content, live classes, tests and performance tracking |
| Educator                          | Creates and manages educational content and tests      | Content upload tools, live class scheduling and student performance analytics      |
| Administrator                     | Manages platform operations                            | User management, content moderation, payment monitoring and system analytics       |
| Payment Gateway (Razorpay/Stripe) | Processes course and subscription payments             | Secure integration, transaction verification and webhook-based confirmation        |
| IT/Support Team                   | Maintains platform infrastructure                      | System monitoring, uptime, data security and issue resolution                      |

### Requirement Elicitation Techniques

| Technique | Purpose                                                                   | Outcome                          |
|-----------|---------------------------------------------------------------------------|----------------------------------|
| Interview | Discuss learning and content-management needs with students and educators | Detailed functional requirements |

| Technique     | Purpose                                                 | Outcome                             |
|---------------|---------------------------------------------------------|-------------------------------------|
| Questionnaire | Collect feedback from multiple students                 | User expectations and preferences   |
| Observation   | Observe how students navigate courses and attempt tests | Usability and workflow requirements |
| User Persona  | Model representative students and educators             | User-centred requirements           |

## Sample Requirement Gathering

Interview Findings:

- Students require easy access to both free and paid educational content.
- Educators need simple tools to upload notes, recorded lectures and schedule live classes.
- Administrators need a single dashboard to manage users, content and payments.
- The test system should support automatic evaluation and instant results.
- The payment system should support subscriptions as well as one-time purchases.

Observation Findings:

- Users prefer simple navigation between courses, tests and live classes.
- Students need a clear indication of free versus paid content.
- Frequently used features such as the dashboard and tests should be easily accessible.
- Test results and rankings should be presented in an understandable format.

Questionnaire Findings:

- Students expect the platform to be available 24x7.
- Access to live classes and recorded lectures is a high priority.
- Users expect secure and reliable payment processing.
- Educators expect detailed analytics on student performance.

## User Personas

| Persona                      | Age | Goal                                              | Pain Point                                               |
|------------------------------|-----|---------------------------------------------------|----------------------------------------------------------|
| Undergraduate Student        | 20  | Access affordable study material and test series  | Learning resources are scattered across multiple sources |
| Working Professional Learner | 28  | Upskill through paid courses and live classes     | Limited time; needs flexible recorded lecture access     |
| Educator/Content Creator     | 30  | Reach students online and track their performance | No unified platform for teaching, testing and analytics  |

## User Stories

| ID  | User Story                                                                                  | Acceptance Criteria                                    |
|-----|---------------------------------------------------------------------------------------------|--------------------------------------------------------|
| US1 | As a student, I want to browse and enroll in courses so that I can access learning content. | Feature works correctly and passes functional testing. |

| ID  | User Story                                                                                                 | Acceptance Criteria                                    |
|-----|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| US2 | As a student, I want to attempt online tests so that I can evaluate my preparation.                        | Feature works correctly and passes functional testing. |
| US3 | As a student, I want to attend live classes so that I can learn directly from educators.                   | Feature works correctly and passes functional testing. |
| US4 | As an educator, I want to upload notes and recorded lectures so that students can access study material.   | Feature works correctly and passes functional testing. |
| US5 | As an administrator, I want to manage payments and subscriptions so that I can ensure secure transactions. | Feature works correctly and passes functional testing. |

## Requirement Gathering Report

The elicitation process identified course management, free and paid content access, online test series, live classes, recorded lectures, payment and subscription management, and performance analytics as the primary requirements. Stakeholder feedback emphasised ease of use, reliable content availability, secure payments, fast and accurate test evaluation, and clear performance tracking for students and educators.

## Observation

Using multiple elicitation techniques improves requirement completeness and reduces ambiguity. Interviews, questionnaires, observation and user personas help capture the different needs and expectations of students, educators and administrators.

## Result

Stakeholders were identified, requirements were elicited using suitable techniques, user stories with acceptance criteria were prepared, and the Requirement Gathering Report for the LMS Platform was successfully completed.